

Gunjan Khut

Boston, MA | (+1) 332-201-9279 | gunjan.khut92@gmail.com | [LinkedIn - gunjankhut](#)

Summary:

OT/Controls and Digital Manufacturing leader with 15+ years of experience spanning industrial automation, robotics, machine vision, and AI/ML-driven analytics for FDA-regulated, 24/7 discrete manufacturing environments. Proven ability to own the full stack from machine-level programming, robotic cell design, and OT network architecture to production-scale computer vision, deep learning, and AI agent pipelines that turn shop-floor telemetry into operational decision support. Track record of translating technical work into business impact: leading a \$1.6M+ annual revenue improvement and 29% lot-processing time reduction as the company's first certified Six Sigma Greenbelt and cutting equipment downtime by 98% through targeted process redesign. Experienced across the manufacturing technology lifecycle, controls infrastructure, cybersecurity and remote access, database architecture, and predictive/anomaly-detection analytics with a multidisciplinary academic foundation and an active PhD researching uncertainty-aware anomaly detection for manufacturing OT networks. Six Sigma Greenbelt-certified.

Education:

University of the Cumberland, USA
New England College, USA
University of Maryland, College Park, USA
NMIMS, Mumbai, India
SMEC Automation Pvt. Ltd., India
North Maharashtra University, India
K J Somaiya Polytechnic, India

Ph.D. in Information Technology (Data Science) (Ongoing)
MS. in Data Science & Analytics
M. Eng. in Robotics
MBA in Business Management
T. Diploma in Industrial Automation
B. Eng. in Electronics & Telecommunications
T. Diploma in Industrial Electronics

Skills:

Tools and Software: Python, R, C++, C#, SQL, vb.net, ROS, PyTorch, OpenCV, TensorFlow, Gen AI, Databricks, Git, Spotfire, Tableau, Power BI, Azure, AWS, GCP, Oracle Analytics Cloud, Spark, AutoCAD, and MS Office Suite.

Industrial Automation expertise: Industrial Automation (Rockwell, Siemens, ABB, GE-Fanuc, Delta), SCADA, MES, Machine vision and sensor systems, Robotics, Motion Control, IT/OT Network integration and Security.

Work Experience:

Senior Staff Controls Engineer, Insulet Corporation, Acton, MA

Jul 2020 - Present

- Led implementation of assembly and process equipment, AGVs, and robots (ABB, Fanuc) to optimize manufacturing operations, including full controls infrastructure design: PLC programming, HMIs, machine vision (Keyence, Cognex), and motion control drives.
- Architected controls and OT network infrastructure for high-speed robotic cells, including IO/device selection, power distribution, vision integration, and scalable data exchange with detailed electrical schematics and functional requirement documentation.
- Developed and implemented computer vision and Deep Learning systems for production (real-time object recognition/tracking, camera calibration, 3D reconstruction, ML-based inspection) and led ML/DL anomaly detection using manufacturing telemetry and sensor streams to improve equipment reliability and reduce false-positive alerts.
- Owned manufacturing SQL Server databases and custom VB.NET production support applications end-to-end, including a company-wide SQL Server upgrade across manufacturing equipment and Data Acquisition Servers; drove query optimization and architecture improvements to scale 24/7 pod-line operations.
- Built AI agents and ML pipelines integrating heterogeneous manufacturing and business data sources into a unified analytics platform, enabling cross-functional, deep-learning-driven operational decision support.
- Designed and deployed operational dashboards, translating shopfloor and OT data into actionable insights for engineering and production stakeholders.
- Architected and deployed end-to-end poke-yoke solutions for Omnipod packaging/labeling at USAM, and improved end-to-end functional final test workflows through checksum validation and secure workflow hardening, enabling USAM to produce clinical builds for all approved product types with full component-level traceability.
- Implemented secure, auditable remote access solutions for production equipment, enabling centralized data collection and vendor support while maintaining security in 24/7 environments.
- Led cross-domain root cause analysis across controls, network, and data layers, and owned OT system lifecycle activities (upgrades, validation, change-controlled deployments) to sustain high availability in continuous manufacturing.
- Led the OP5 NPI Pod Program deployment, expanding production from 2 to 7 product types (G6/G7, G6/Libre 2, Gov Pod G6/G7, Atlas, Orion, Moonshot, OTA), including DCS logic development and multi-product line integration at USAM.

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- Integrated a real-time vision inspection station for clutch spring presence/orientation and upgraded existing vision inspection, minimizing faults and supporting a daily production output of 140K+ parts.
- Developed an automated Auto TCP correction solution for ABB robots to continuously monitor and adjust teach points, cutting reteaching requirements by an estimated 12-14 hours per week.
- Redesigned conveyor HMI logic, reducing associated downtime from 4 hours to 10 minutes (a 98% improvement).
- Completed two Greenbelt projects and one Six Sigma project as the company's first certified Greenbelt Six Sigma professional, eliminating labeling NCMRs, reducing scrap costs by \$450,698 annually, cutting lot processing time by 29%, and driving \$1,666,536 in additional annual revenue.

Business Intelligence Developer, Sylger Corporation, Washington, DC

Jun 2020 - Sep 2020

- Developed scalable ML solutions for CSOSA (Court Services and Offender Supervision Agency) throughout the entire project lifecycle, including data extraction/preparation, result documentation on Azure Big Data platforms. Processed and analyzed a dataset of 50M offender records, achieving accuracy of 98%, resulting in improved model performance.

Software Engineer Intern, digiBlitz, Herndon, Virginia

Feb 2020 - Jun 2020

- Conducted large-scale data training and validation for various DNNs implemented in TensorFlow. Handling datasets of over 100,000 samples with an accuracy enhancement of 15% on average.
- Designed and programmed robotic arms, resulting in a 25% increased precision on robot control and device controllers.

Research Assistant, University of Maryland, College Park,

May 2018 - Dec 2019

- Led the development of scalable ML solutions, including facial recognition, achieving 92% recognition accuracy.
- Designed and executed various SQL queries and functions, meeting demands with a 98% accuracy rate in query results.
- Implemented ML algorithms using Azure for detailed analytics, resulting in a 75% improvement in model performance.
- Updated Python scripts to match training data with the AWS Cloud Search database for document classification.

Automation Engineer, P. G. Drive, Mumbai, India

July 2015 - Dec 2017

- Performed testing, commissioning, and maintenance of electrical panels, PLCs, VFDs, and HMIs for various projects.
- Innovated and upgraded a manual solar tracker to a sensor-based model for automation.
- Developed control logic, AutoCAD drawings, Technical specification documents, power & control wiring, specifications, and inspection of low-medium power electrical equipment: switchgears, transformers, motor testing.
- Led various field operational projects from design through commissioning, providing after-sales and technical support.
- Designed electrical control panels and successfully commissioned complex solutions involving IT networking and automation components, yielding a 20% enhancement in production efficiency and reduction in manual intervention.
- Prepared instrument data sheet, flow charts, BOM, and erection bill of materials based on the cost-effective solutions.

Maintenance Engineer, JK & PC Textlab Equipments, Mumbai, India

Jun 2014 - Jun 2015

- Installed 15 complex and advanced machines for clothing yarns, resulting in a 40% increase in production capacity.
- Updated the technical documentation database and performed analysis to find 6 process innovations.
- Contributed to the enhancement of the product portfolio by working on special projects for the beaker dyeing machine.
- Diagnosed and troubleshooted system errors, implemented tactical solutions to improve machine efficiency by 15%.
- Calibrated and performed FAT for the instruments, ensuring quality standards and a 100% passing rate.

Electronics Engineer, Speed Control Industries, Mumbai, India

Jun 2012 - Jun 2014

- Configured module software, generated I/O Databases, and translated flowcharts and write-ups to configurations.
- Engaged in system engineering, including designing PCBs, IoT integration, assigning Digital I/O points, preparing BOMs, and wiring diagrams. Prepared comprehensive documentation, which reduced troubleshooting time by 60%.
- Coordinated with contractors and customers during site visits, meetings, and document preparation for successful project implementation with 95% customer satisfaction rate.

Projects:

Designing and Validating a Domain-Adapted Uncertainty-Aware Anomaly Detection Framework for Discrete Manufacturing OT Networks (Ongoing PhD Research)

- Developing a DL-based anomaly detector with Bayesian MC-dropout uncertainty quantification and conformal prediction to produce confidence-calibrated alerts, reporting Expected Calibration Error (ECE) against deterministic baselines.
- Evaluating cross-domain transferability against continuous-process ICS benchmarks (SWaT, WADI, HAI) using de-identified manufacturing OT telemetry, with red-team attack labeling aligned to MITRE ATT&CK for ICS.

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Homelab: End-to-End IT/OT Infrastructure

- Architected a full-stack lab environment network architecture, firewalls, VPN, hypervisors, virtualized servers, NAS storage, and segmented networks for hands-on OT, cybersecurity, and AI/ML experimentation.

SSA-CRM: Design, Procure, Print & Packaging CRM

- Designed and built a full-stack CRM (React, Node.js, Express, Prisma, PostgreSQL) covering clients, jobs, approvals, payments, and role-based permissions, with JWT auth and an admin-configurable permission matrix.

Anomaly Detection in Poly Etch Process, Analog Devices

- Implemented refined data preprocessing with time slicing and interpolation for uniformity, coupled with an advanced anomaly detection system utilizing Autoencoders and K-Means-based Univariate Isolation Forest models, resulting in a validated model with 91% prediction rate for anomaly detection and a robust framework for real-time operational monitoring.

Tableau Dashboards:

- Developed and deployed 5 interactive dashboards and visualizations using Tableau, enhancing visibility into quality metrics and reducing reporting time by 50%. Established and maintained best practices, including data sourcing, calculations, validation, and reporting.

Path Planning Algorithm on AGV

- Planning routes and navigating obstacles using LiDAR from a start point to end point in a known simulated environment.
- Used g-mapping technique using SLAM and developed a binary map of the environment to feed to the RRT algorithm.

Emergency Vehicle Detection using TensorFlow API (Winner of Northrop Grumman Challenge)

- Worked on an emergency vehicle detection system in low visibility conditions, detected vehicles with accuracy of more than 92% on images/videos using transfer learning and faster RCNN model.
- Involved selection of an appropriate CNN model and data augmentation.

A* path planning for a differential drive robot

- Used A* algorithm to find a path between two points in a 2D space, obtained the C-Space map from gmap by extracting from the provided rrlab.sdf file then converted to a .png file; the map was processed with OpenCV, setting 1's as obstacles and 0's as free space, and the robot's known radius was used to pad 1's around obstacles for point robot consideration.

ARIAC (Agile Robotics for Industrial Automation Competition)

- Established control between collaborative robots (UR10 and AGV) to fulfil the orders given to the competition environment and moved parts from assembly bins to AGV's. MoveIt & Gazebo are the ROS plugins used.
- Improvised the system by adding contingencies for Part Drop and Important Order First. (C++, ROS)

SunGanak – A solar tool

- A data science-powered web-software giving a generation report of various parameters such as Return on Investment (ROI), carbon footprints, and estimated generation of a solar PV Plant.

GPS Vault with Fingerprint Scanner & Keypad Lock (Winner of Best innovative project)

- Engineered a sequential multi-layered security vault that was operational only when it fulfilled the following security combinations: Coordinate specific location, Biometric recognition, Encrypted 4-digit pin.

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Some Projects at Insulet:

- **Greenbelt and Six Sigma Certification Projects:** Successfully completed two Greenbelt projects and one Six Sigma project, becoming the company's first certified Greenbelt Six Sigma professional, with initiatives that eliminated labeling NCMRs at C95, reduced scrap costs by \$450,698 annually, decreased lot processing time by 29%, and increased production capacity, resulting in an additional \$1,666,536 in annual revenue.
- **New Product Introduction:** Support the deployment of the OP5 NPI Pod Program, exceeding expectations by expanding from 2 to 7 product types, including G6/G7, G6/Libre 2, Gov Pod G6/G7, Atlas, Orion, Moonshot, and OTA. Updated USAM production lines to accommodate new parts and products, developed logic for DCS requirements, and expanded support for integrating multiple product types in production.
- **Design Review and Implementation:** Successfully supported and executed the back half L4 design review, installation, and startup, including cell installation, comprehensive testing (FAT/SAT/IO/OQ/PQ), and all necessary tasks, resulting in fully operational production cells that significantly enhanced manufacturing capacity and efficiency.
- **Data Analytics Dashboard:** Developed a comprehensive data collection logic for station cycle times and assisted the Data Analytics team in creating a transparent analytics dashboard, empowering the team to gain insights into station cycle timings, rate per minute, and machine performance, enabling data-driven decision-making and improving operational efficiency across production processes.
- **Data Tracking:** Developed systems to handle duplicate tray serial numbers, implemented tray tracking for correct product traceability, and addressed tray stacking and serialization issues in operations. Improved tray tracking accuracy, minimizing serialization errors and enhancing product traceability across production lines.
- **Vision System and Server Enhancements:** Addressed server disconnection issues, expanded vision server storage capacity, and deployed a Python application for machine tag logging. Resolved system bottlenecks to improve the continuity of critical production systems. Improved system stability and reliability, reducing technical disruptions and enhancing the overall performance of critical production systems.
- **Auto TCP Correction for ABB Robots (Test Station):** Developing an automated solution to continuously monitor and adjust teach points on ABB Robots, ensuring programmed points remain accurate over time. This innovation will significantly reduce reteaching requirements, saving a remarkable 12-14 hours per week.
- **Vision System Integration and Upgrade:** Integrated a vision system as a new station, enabling real-time checks for clutch spring presence and orientation. Also, improved existing vision inspection. This minimized faults and optimized machine performance, achieving an impressive daily production output of 140K+ parts.
- **Laser Marker Program Upgrades:** Elevated operational efficiency by upgrading laser marker programs for a diverse range of product types, including 8 versions of Inspire and OP5 G6/G7. Refined existing programs to improve performance across multiple product categories.
- **Green Belt Quality Project: Validated Parameters Digitization Initiative (eDHR Component):** Led validated parameters digitization initiative as an eDHR component, digitizing machine parameters via HMI/VP Buddy to replace manual recording with automated, audit-ready data capture for FDA-regulated production. This data-driven approach facilitated enhanced quality monitoring, contributing to overall process optimization.
- **Remote Machine Access and Data Analysis:** Successfully added remote access capabilities to gather machine data for a centralized database. Using data analysis techniques will help refine OEE and MOEE, supporting continuous improvement.
- **Enhanced Robot Control and Performance Optimization:** Corrected and optimized Robot 1's unknown position, improvements in overall cell performance by minimizing robot reteaches and homing requirements.
- **Conveyors HMI Redesign:** Improved HMI. Reducing downtime from 4 hrs. to 10 mins. i.e., 98% improvement
- **Test Method Validations Protocols:** Performed various data collection tests to gather information. Utilized observations to formulate a comprehensive protocol for aTMV for OmniLine (Dash & OP5). Successfully executed the protocol during the Line Validation of the new Omniline - PDM manufacturing.
- **Process Improvement and Technology Integration:** Transformed how things are done, made things work better, and used technology to improve how we make things and measure their quality in various projects.

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- **Functional Final Test Workflow Improvement:** Identified vulnerabilities in end-to-end functional final test processes; implemented checksum checks, process upgrades, and secure gap closure to improve test integrity and production reliability.
- **Manufacturing Database & VB Application Ownership:** Own and enhance custom VB.NET applications and manufacturing databases supporting production; drive reliability, scalability, and ongoing solution development for critical shop-floor systems.
- **Manufacturing Database Optimization & Architecture Improvements:** Led query optimization, stored procedure improvements, and robust database architecture enhancements on manufacturing equipment; analyzed system logs and prepared reports for DBA and IT stakeholders to drive performance, reliability, and scalability improvements.
- **Rework & Reclaim Process Improvement:** Improved rework and reclaim workflows by addressing process gaps and updating controls to ensure end-to-end traceability is maintained throughout recovery, reuse, and production operations.
- **AI/ML/Deep Learning Integration for Smart Manufacturing:** Integrated AI, ML, and deep learning across production workflows, computer vision inspection, predictive maintenance models, anomaly detection, and AI agents connecting OT sensor data, historian, and business systems into actionable manufacturing intelligence.
- **Multi-Source Data Integration Agents:** Built an AI agent and ML-driven pipelines to consolidate data from multiple sources into a common platform and deliver actionable manufacturing and quality insights.
- **SQL Server Manufacturing Upgrade:** Owned end-to-end upgrade for manufacturing databases, Data Acquisition Servers, and production support applications; coordinated validation, testing, and production bring-up across USAM operations.
- **USAM Omnipod Packaging & Labeling Poke-Yoke:** Designed and implemented end-to-end error-proofing for packaging and labeling processes, preventing labeling defects through automated checks and process controls.
- **Clinical Build Production & Component Traceability:** Enabled clinical build manufacturing for all approved product types at USAM with full component-level traceability across the production lifecycle.
- **Robotic Cell OT Architecture & Network Design:** Led end-to-end OT design for robotic manufacturing cells, controls infrastructure, network architecture, vision system integration, and documentation of the sequence of operation and functional requirements for high-precision discrete manufacturing.
- **Secure Remote Access & Centralized OT Data Collection:** Deployed secure remote access and centralized machine data pipelines to manufacturing databases, supporting OEE/MOEE analysis, continuous improvement, and vendor troubleshooting without disrupting production.
- **OT Telemetry Predictive Monitoring & Anomaly Detection:** Built ML and deep learning monitoring and anomaly detection frameworks using production telemetry and sensor data (including Autoencoder, Isolation Forest, and neural network approaches) to establish behavioral baselines and improve process reliability.
- **Manufacturing Operations Analytics Dashboards:** Developed transparent analytics dashboards for station cycle times, rate per minute, and machine performance, empowering data-driven decision-making across USAM production processes.
- **OT System Lifecycle & Infrastructure Upgrades:** Managed lifecycle upgrades across SQL Server databases, vision servers, and Data Acquisition infrastructure, expanding storage, resolving bottlenecks, and validating systems for production readiness post-upgrade.